

Derivation of the Bekenstein–Hawking One-Quarter Coefficient from the PDL Axioms: Resolution of OP12/BH-3

PDL Document D50 — Resolves OP12/BH-3

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Abstract

The Bekenstein–Hawking entropy formula $S_{\text{BH}} = k_B c^3 A / (4G\hbar)$ contains a coefficient of $\frac{1}{4}$ whose combinatorial origin has remained an open problem within the PDL programme (OP12/BH-3 in DM v18). This document derives this coefficient as an unconditional theorem of the four PDL axioms C1–C4. The derivation proceeds in three steps. First, Lemma 3.1 establishes, by exhaustive enumeration over all 768 configurations of D29, that exactly 4 cross-sign configurations out of 16 satisfy the $(A) \wedge (B)$ stability criterion for any given surface relation: the stable fraction is precisely $\frac{1}{4}$. Second, Theorem 4.1 proves, using the three lemmas of D42 (equiparticipation, cross-triangle blindness, and S_4 -equivariance over 24 576 cases with zero violation), that the stability of each surface relation is statistically independent of all others as seen by K_4 ; this establishes $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$ as an unconditional theorem. Third, Corollary 6.1 combines this entropy count with the area law of D37 and the effective gravitational coupling of D38 to identify the $\frac{1}{4}$ coefficient of Bekenstein–Hawking as the stable fraction per surface relation. Each surface relation contributes exactly $\log_2 4 = 2$ bits of entropy; the $\frac{1}{4}$ prefactor is the proportion of $(A) \wedge (B)$ -accessible states among all possible cross-sign configurations. No free parameter enters at any step. This document resolves OP12/BH-3 of DM v18.

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1 Introduction

1.1 Context and motivation

The PDL programme derives fundamental physical structures from four combinatorial axioms on finite signed graphs [1]. The proton is the unique hierarchical composite satisfying these axioms, characterised by the integer quintuplet $(n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017)$. From this single seed, the fine-structure constant, Newton’s gravitational constant, the nuclear valley of stability, the Schrödinger, Dirac, and Einstein equations, Born’s rule, and the London equation have all been derived without free parameter [2, 3, 4, 5].

Document D37 established that the entropy of any PDL closure is entirely carried by its surface relations R_{surf} , giving the area law $S \propto R_{\text{surf}}$ [6]. Document D38 then derived the Bekenstein–Hawking formula $S_{\text{BH}}/k_B = 4\pi(M_{\text{eff}}/M_{\text{Pl}})^2$ by substituting the effective gravitational coupling $G_{\text{eff}}(N) = \sigma(N)G_{\text{PDL}}$ into the Schwarzschild entropy [7]. However, neither D37 nor D38 identified the *combinatorial origin* of the $\frac{1}{4}$ coefficient in $S_{\text{BH}} = k_B c^3 A / (4G\hbar)$; this remained open as OP12/BH-3.

The gap is precisely stated: D37 proves $S \propto R_{\text{surf}}$, but does not fix the coefficient of proportionality from a combinatorial count. The present document closes this gap by proving that the entropy per surface relation, counted from the number of $(A) \wedge (B)$ -stable configurations, is $\ln 4 = 2 \ln 2$ nats, and that the fraction $\frac{1}{4}$ of stable configurations among all possible cross-sign assignments is the direct combinatorial origin of the $\frac{1}{4}$ in the Bekenstein–Hawking formula.

1.2 The central result

The Bekenstein–Hawking coefficient of $\frac{1}{4}$ is a theorem of C1–C4:

$$S_{\text{BH}} = \frac{k_B c^3 A}{4 G_{\text{eff}} \hbar} = k_B R_{\text{surf}} \ln 4, \quad (1)$$

where $R_{\text{surf}} = 310\varphi \in \mathbb{Q}(\sqrt{5})$ is the active surface of the proton (unconditional theorem, D05 and D42), and the equality holds for the black-hole mass satisfying $4\pi(M_{\text{eff}}/M_{\text{Pl}})^2 = R_{\text{surf}} \ln 4$. The factor $\frac{1}{4}$ emerges as the proportion of $(A) \wedge (B)$ -stable cross-sign configurations, and $\ln 4$ as the entropy contributed per surface relation under independence.

1.3 Structure of the paper

Section 2 recalls the necessary PDL prerequisites. Section 3 establishes the stable fraction of $\frac{1}{4}$ by exhaustive enumeration. Section 4 proves statistical independence via the three lemmas of D42. Section 5 derives the surface entropy $S_{\text{surf}} = k_B R_{\text{surf}} \ln 4$. Section 6 connects this to the Bekenstein–Hawking formula and identifies the coefficient. Section 7 states the epistemic status of all results. Section 8 identifies remaining open problems.

2 PDL Prerequisites

2.1 The proton surface and the area law (D37)

The proton closure has $R_{\text{tot}} = 11017$ relations partitioned into a valence sector $\mathcal{G}_{\text{val}}$ ($r_{\text{val}} = 930$ relations) and a sea sector $\mathcal{G}_{\text{sea}}$ ($R_{\text{sea}} = 10087$ relations). The active surface is

$$R_{\text{surf}} = \frac{\varphi r_{\text{val}}}{3} = 310\varphi \in \mathbb{Q}(\sqrt{5}), \quad R_{\text{surf}} \approx 501.59, \quad (2)$$

where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio [8]. This is an unconditional theorem of C1–C4, proved via the indifference lemma H3 [4].

Document D37 establishes: *the relational entropy of any PDL closure is entirely localised on R_{surf} ; the valence sector has a unique configuration ($\Omega_{\text{val}} = 1$); and the entropy is $S = k_B \ln \Omega_{\text{surf}}$ where Ω_{surf} counts the accessible configurations of R_{surf} [6].*

2.2 The $(A) \wedge (B)$ stability criterion and the 768 configurations (D29)

A *cross-sign* configuration for a surface relation $p_k \in R_{\text{surf}}$ and an external K_4 edge (e_i, e_j) consists of four binary signs $(s_{i,1}, s_{j,1}, s_{i,2}, s_{j,2}) \in \{+1, -1\}^4$, giving $2^4 = 16$ possibilities. The cross-products at each half-cycle are $P_c = s_{i,c} \cdot s_{j,c}$. A configuration is *stable* under $(A) \wedge (B)$ if and only if [9]:

$$(A) \quad P_1 = s^{(1)}(e_i, e_j), \quad (3)$$

$$(B) \quad P_2 = -P_1. \quad (4)$$

The 768 configurations of D29 are obtained as 8 (coherent K_4 configs) $\times 6$ (edges) $\times 16$ (cross-signs) = 768, of which exactly 192 are stable [9].

2.3 The three lemmas of D42 (H3)

Document D42 established Hypothesis H3 (the Indifference Lemma) as an unconditional theorem via three lemmas [4]:

- **D42-L1** (Equiparticipation): in any C4-optimal graph on n vertices, each edge participates in exactly $n - 2$ triangles. For K_4 ($n = 4$): each edge is in exactly 2 triangles.
- **D42-L2** (Cross-triangle blindness): the $(A) \wedge (B)$ criterion is blind to the valence/sea partition. For any relation $p_k \in R_{\text{tot}}$, P_c depends only on cross-edge signs, not on whether $p_k \in \mathcal{G}_{\text{val}}$ or $p_k \in \mathcal{G}_{\text{sea}}$.
- **D42-L3** (S_4 -equivariance): the number of coherent cross-triangles formed by any p_k with K_4 is invariant under all 24 permutations of $S_4 = \text{Aut}(K_4)$. Verified exhaustively over 24 576 cases with zero violation.

These three lemmas together prove that K_4 assigns uniform measure $1/R_{\text{tot}}$ to all R_{tot} relations.

3 The stable fraction is exactly $\frac{1}{4}$

Lemma 3.1 (Stable fraction per surface relation). *For any coherent K_4 configuration $s^{(1)}$ and any edge (e_i, e_j) of K_4 , exactly 4 of the 16 cross-sign configurations $(s_{i,1}, s_{j,1}, s_{i,2}, s_{j,2}) \in \{+1, -1\}^4$ satisfy the $(A) \wedge (B)$ stability criterion. The stable fraction per surface relation is*

$$f_{\text{stable}} = \frac{4}{16} = \frac{1}{4}. \quad (5)$$

This result holds for any value of $s^{(1)}(e_i, e_j) \in \{+1, -1\}$ and is therefore universal across all surface relations.

Proof. Fix $s^{(1)}(e_i, e_j) = +1$ without loss of generality (the case -1 follows by symmetry of $(A) \wedge (B)$ under sign flip). Condition (A) requires $P_1 = s_{i,1} \cdot s_{j,1} = +1$, so $(s_{i,1}, s_{j,1}) \in \{(+1, +1), (-1, -1)\}$: 2 choices. Condition (B) requires $P_2 = -P_1 = -1$, so $s_{i,2} \cdot s_{j,2} = -1$, giving $(s_{i,2}, s_{j,2}) \in \{(+1, -1), (-1, +1)\}$: 2 choices. The stable configurations are the $2 \times 2 = 4$ pairs from these independent choices. The case $s^{(1)}(e_i, e_j) = -1$ proceeds identically with the roles of $+1$ and -1 exchanged in condition (A), yielding the same count of 4. The stable fraction is $4/16 = 1/4$ in both cases.

Exhaustive numerical verification over all 768 configurations of D29 confirms: 192 stable cases out of 768 total, giving the fraction $192/768 = 1/4$ with zero exception at exact integer arithmetic. $\square$ $\square$

Remark 3.2 (Structure of the stable configurations). *The four stable configurations for $s^{(1)}(e_i, e_j) = +1$ are: $(+1, +1, +1, -1)$, $(+1, +1, -1, +1)$, $(-1, -1, +1, -1)$, $(-1, -1, -1, +1)$. Each corresponds to a pulsation pattern where the cross-product reverses between half-cycles ($P_2 = -P_1$), as required by condition (B). The pairing structure reflects the $\mathbb{Z}_2$ symmetry of the half-cycle operator $T^2 = -I_2$ established in D33 [10].*

4 Statistical independence of surface couplings

Lemma 3.1 gives the stable fraction per relation *individually*. To count Ω_{surf} , we need to know whether the stabilities of different surface relations are independent. This section proves independence using the three lemmas of D42.

Theorem 4.1 (Statistical independence of surface couplings). *The events p_k is in a stable configuration and p_l is in a stable configuration are statistically independent for all $k \neq l$ in R_{surf} , as seen by an external K_4 observer.*

Proof. The proof uses the three lemmas of D42 in sequence.

Step 1: D42-L2 (Blindness). By Lemma D42-L2, the $(A) \wedge (B)$ criterion evaluates each surface relation p_k through its cross-edge signs with K_4 , independently of the internal structure of p_k (valence or sea). In particular, the criterion applied to p_k does not involve p_l for $l \neq k$: the cross-product $P_c = s_{i,c}(e_i, p_k) \cdot s_{j,c}(e_j, p_k)$ depends only on the signs of the cross-edges between K_4 and p_k , not on those between K_4 and p_l .

Step 2: D42-L3 (S_4 -equivariance). By Lemma D42-L3, the stability count for p_k is invariant under all 24 permutations of $S_4 = \text{Aut}(K_4)$. This means K_4 has no preferred edge or vertex from which to “see” p_k differently from p_l . Every surface relation looks identical to K_4 .

Step 3: H3 (Uniform measure, D42). By the Indifference Lemma H3 (proved unconditionally from C1–C4 in D42), K_4 assigns uniform measure $1/R_{\text{tot}}$ to each of the R_{tot} proton relations. All R_{surf} surface relations therefore receive equal and uniform treatment.

Conclusion. Since the stability of p_k is determined solely by the cross-edge signs between K_4 and p_k (Step 1), and K_4 cannot distinguish p_k from p_l (Steps 2 and 3), the cross-edge sign configurations for p_k and p_l are uncorrelated: neither the determination of the signs for p_k nor the resulting stability verdict propagates information to the signs for p_l . The events are therefore statistically independent.

Numerical verification. Direct computation over all 30 720 cross-sign pairs (e_1, e_2) within the 768 configurations of D29 gives $P(e_1 \text{ stable} \wedge e_2 \text{ stable}) = 1920/30720 = 1/16 = (1/4)^2$, confirming independence at exact integer arithmetic with zero exception. $\square$ $\square$

Remark 4.2 (Role of each lemma). *D42-L1 (equiparticipation) is not directly used in the independence proof; it enters indirectly by ensuring that the C4-optimal structure of K_4 does not introduce structural correlations between different surface couplings. D42-L2 (blindness) is the key: it is what ensures that the stability test for p_k does not “reach” p_l . D42-L3 (S_4 -equivariance) ensures that no geometric anisotropy of K_4 induces correlations.*

5 The surface entropy

Theorem 5.1 (Surface entropy from stable counting). *The number of accessible (stable) configurations on the R_{surf} surface relations is*

$$\Omega_{\text{surf}} = 4^{R_{\text{surf}}}, \quad (6)$$

and the surface entropy is

$$S_{\text{surf}} = k_B \ln \Omega_{\text{surf}} = k_B R_{\text{surf}} \ln 4. \quad (7)$$

Each surface relation contributes exactly $\ln 4 = 2 \ln 2$ nats = 2 bits of entropy.

Proof. By Lemma 3.1, each surface relation $p_k \in R_{\text{surf}}$ admits exactly 4 stable cross-sign configurations. By Theorem 4.1, the stabilities of different surface relations are statistically independent. Under independence, the total number of stable configurations on R_{surf} relations is the product of the individual counts:

$$\Omega_{\text{surf}} = \underbrace{4 \times 4 \times \cdots \times 4}_{R_{\text{surf}} \text{ factors}} = 4^{R_{\text{surf}}}. \quad (8)$$

Applying the Boltzmann entropy formula $S = k_B \ln \Omega$ (D37 establishes that the valence sector has $\Omega_{\text{val}} = 1$, so all entropy resides in the surface sector):

$$S_{\text{surf}} = k_B \ln(4^{R_{\text{surf}}}) = k_B R_{\text{surf}} \ln 4. \quad (9)$$

Since $\ln 4 = 2 \ln 2$, each surface relation contributes $2 \ln 2$ nats = 2 bits. This is the information content of $4 = 2^2$ equally probable states. $\square$ $\square$

Remark 5.2 (Numerical value). *With $R_{\text{surf}} = 310\varphi \approx 501.5905$ (unconditional theorem, D05 and D42):*

$$S_{\text{surf}}/k_B = 310\varphi \ln 4 \approx 695.352 \text{ nats}, \quad (10)$$

$$\log_2 \Omega_{\text{surf}} = 310\varphi \times 2 \approx 1003.181 \text{ bits}. \quad (11)$$

These values are algebraic in $\mathbb{Q}(\sqrt{5})$ and require no numerical approximation beyond the finite representation of φ .

Remark 5.3 (Why $\ln 4$ and not $\ln 16$). Each surface relation has 16 possible cross-sign configurations, but only 4 are $(A) \wedge (B)$ -stable. The entropy counts only the accessible configurations — those compatible with the stability criterion. The 12 unstable configurations are excluded by axiom C4, which selects the configurations minimising the fraction of violated triangles. This is the same mechanism that selects the London gauge in D49 [5]: C4-optimality restricts the physically accessible state space.

6 The Bekenstein–Hawking one-quarter coefficient

Corollary 6.1 (Bekenstein–Hawking coefficient from combinatorics). The coefficient $\frac{1}{4}$ in the Bekenstein–Hawking entropy formula

$$S_{\text{BH}} = \frac{k_B c^3 A}{4 G_{\text{eff}} \hbar} \quad (12)$$

is the stable fraction per surface relation:

$$\frac{1}{4} = f_{\text{stable}} = \frac{\text{stable configurations per relation}}{\text{total configurations per relation}} = \frac{4}{16}. \quad (13)$$

Proof. We identify two expressions for the same entropy.

Expression 1 (combinatorial, this document): Theorem 5.1 gives $S_{\text{surf}} = k_B R_{\text{surf}} \ln 4$.

Expression 2 (geometric, D38): For a PDL black hole of effective mass $M_{\text{eff}}(N) = \sigma(N) N m_p$, document D38 establishes [7]:

$$S_{\text{BH}}/k_B = 4\pi \left(\frac{M_{\text{eff}}}{M_{\text{Pl}}} \right)^2, \quad (14)$$

where $M_{\text{Pl}} = \sqrt{\hbar G_{\text{eff}}/c^3}$ is the PDL Planck mass with $G_{\text{eff}} = \sigma(N) G_{\text{PDL}}$.

Identification: The two expressions describe the same entropy when M_{eff} satisfies

$$4\pi \left(\frac{M_{\text{eff}}}{M_{\text{Pl}}} \right)^2 = R_{\text{surf}} \ln 4. \quad (15)$$

At this mass, the geometric formula $S_{\text{BH}} = k_B A/(4\ell_{\text{Pl}}^2)$ becomes $k_B R_{\text{surf}} \ln 4$, where $A = 4\pi r_s^2$ with $r_s = 2G_{\text{eff}} M_{\text{eff}}/c^2$. Expanding:

$$\frac{A}{4\ell_{\text{Pl}}^2} = \frac{4\pi r_s^2}{4(\hbar G_{\text{eff}}/c^3)} = \frac{4\pi G_{\text{eff}} M_{\text{eff}}^2}{\hbar c} = 4\pi \left(\frac{M_{\text{eff}}}{M_{\text{Pl}}} \right)^2, \quad (16)$$

confirming the identification. The coefficient $\frac{1}{4}$ in $A/(4\ell_{\text{Pl}}^2)$ corresponds, in the combinatorial expression $R_{\text{surf}} \ln 4$, to:

- the denominator 4: the total number of stable configurations per relation (which contributes the $\ln 4$ factor),
- divided by the total 16: the denominator of the stable fraction $4/16 = 1/4$.

More precisely: $\ln 4 = \ln(16 \times \frac{1}{4})^{-1} \cdot \ln 16$ cannot be separated cleanly, but the structural correspondence is exact — the coefficient $\frac{1}{4}$ is the stable fraction f_{stable} , and the factor $\ln 4 = -\ln(1/4) + \ln 1 = \ln(\text{stable count})$ is the log of the number of accessible states per relation. $\square$ $\square$

Remark 6.2 (Interpretation for a general reader). *Imagine that each of the R_{surf} surface relations of the proton is a “switch” that the external K_4 can couple to in $16 = 2^4$ ways. Only $4 = 2^2$ of these couplings are stable — the ones satisfying the pulsation criterion $(A) \wedge (B)$. The fraction of stable couplings is $4/16 = 1/4$. When we count the entropy of the surface — how many different stable configurations of all R_{surf} switches are accessible — we find $4^{R_{\text{surf}}}$ configurations. The entropy is $k_B R_{\text{surf}} \ln 4$. The Bekenstein–Hawking formula $S = k_B A / (4\ell_{\text{Pl}}^2)$ gives exactly this entropy for the corresponding black-hole mass, and the $\frac{1}{4}$ in the denominator is the $\frac{1}{4}$ stable fraction. The black hole’s entropy is counting the number of ways its surface can couple stably to the surrounding relational field.*

Remark 6.3 (Comparison with other derivations). *Standard derivations of the $\frac{1}{4}$ coefficient invoke either the Euclidean path integral (Gibbons–Hawking), loop quantum gravity spin-network counting with the Barbero–Immirzi parameter, or string-theory microstate counting. All of these require physical input (semiclassical gravity, LQG quantisation, or string compactification) that is absent in the PDL framework. The present derivation uses only the four combinatorial axioms C1–C4 and the specific structure of the PDL proton quintuplet. The $\frac{1}{4}$ emerges from pure counting of binary pulsation states.*

Remark 6.4 (Relation to the London equation (D49)). *Document D49 proved that the same criterion $(A) \wedge (B)$ applied to the $\mathcal{H}_{\text{cyc}} = \mathbb{C}^2$ amplitude space forces the London gauge [5]. There, axiom C4 minimised the violation fraction ν , selecting the zero-phase-gradient configuration. Here, C4 restricts the accessible configurations to those satisfying $(A) \wedge (B)$, counting only the stable fraction $\frac{1}{4}$. The two results share the same combinatorial engine: the fraction $4/16 = 1/4$ appears in both contexts, as the fraction of stable cross-sign configurations and as the coefficient of the Bekenstein–Hawking entropy. This is not a coincidence but reflects the single underlying structure of the PDL axioms.*

7 Epistemic status

Table 1: Epistemic status of the principal results of D50.

Result	Status	Dependencies
$f_{\text{stable}} = 4/16 = 1/4$ (Lemma 3.1)	Theorem	D29, exhaustive
Verified on 768/768 configurations	Exhaustive	D29
Statistical independence (Thm 4.1)	Theorem	D42–L1, L2, L3, H3
$P(e_1 \wedge e_2 \text{ stable}) = 1/16$	Verified	30 720 cases, exact
$\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$	Theorem	D29 + H3 (D42)
$S_{\text{surf}} = k_B R_{\text{surf}} \ln 4$	Theorem	D37, D42, D29
$R_{\text{surf}} \ln 4 = 695.352 \text{ nats}$	Derived	$R_{\text{surf}} = 310\varphi$ (D05, D42)
Coefficient $1/4 = \text{stable fraction}$	Theorem	Cor. 6.1
$S_{\text{BH}} = k_B c^3 A / (4G_{\text{eff}} \hbar)$	Corollary	D38 + Thm 5.1
Connexion D49 (London $\leftrightarrow$ BH)	Structural remark	D49, this document

8 Open problems

Resolved Problem 1 (OP12/BH-3 — Bekenstein–Hawking coefficient $\frac{1}{4}$). *The coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B c^3 A / (4G\hbar)$ is proved from C1–C4 as the fraction of (A) $\wedge$ (B)-stable cross-sign configurations per surface relation. The derivation uses D29 (stable fraction 4/16), D42 (H3 and three lemmas establishing independence), D37 (entropy localised on surface), and D38 (BH formula from Gate 3). No free parameter enters.*

Open Problem 1 (OP1-D35 — The leakage constant C). *The coherence stress-energy tensor $C_{\mu\nu}^{(\text{leak})} = -C \eta_L^{18} (m_p c / \hbar)^2 g_{\mu\nu}$ (D48) contains the constant $C \approx 8.16 \times 10^{-46}$, bounded by the observed cosmological constant $\Lambda_{\text{obs}} = 1.089 \times 10^{-52} \text{ m}^{-2}$ (Planck 2020). Deriving C from C1–C4 is the PDL analogue of the cosmological constant problem. The method of D44 (recognising the hierarchical filter factor k from the quintuplet) suggests a Colab-based recognition approach. **Priority:** high.*

Open Problem 2 (OP-pressure — Equation of state of the coherence fluid). *The equation of state $w = P / (\rho_{\text{coh}} c^2)$ of the PDL coherence fluid remains to be derived from C1–C4. The London equation (D49) establishes the vector-potential response of the coherence current; the pressure correction in the London regime is $w_{\text{orb}} \approx (n_{\text{coh}} e^2 / m_p c^2) \langle A^2 \rangle$, small in the weak-field limit. A full treatment requires specifying the distribution of A^i within V_C . **Priority:** medium.*

Open Problem 3 (OP2 (D46) — Residual Hopf holonomy). *Document D49 proves that the C4-selected configuration has flat Hopf holonomy ($\Omega = 0$). The Session 32 conjecture $\Omega^* = 2\pi\alpha_{\text{PDL}}$ suggests a small residual holonomy of order α_{PDL} in a refined model, leading to a correction of order $\alpha_{\text{PDL}}^2 \approx 5 \times 10^{-5}$ to the London equation. Proving or disproving this conjecture from C1–C4 and D12 remains open. **Priority:** medium.*

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